

YUHENG DING

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EDUCATION

Carnegie Mellon University

Master of Science in Intelligent Information Systems (ML/NLP @ SCS LTI)

Pittsburgh, PA
December 2026

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science, Bachelor of Science in Applied Mathematics

Los Angeles, CA
June 2025

GPA: 3.88/4.0, Cum Laude

Relevant Courses: Algorithm and Complexity, Database Systems, Data Structure, Object Oriented Design, Operating Systems, Computer Network, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

EXPERIENCE

VisionX LLC

Full-Stack Software Engineer Intern

San Jose, CA (Remote)
July 2023 - October 2023

- Collaborated with a team of four engineers and contributed to software development of a full-stack e-commerce platform, with a focus on promoting a healthy lifestyle and fostering environmental sustainability.
- Engineered a product page UI template using **Python Django** and **MongoDB**, empowering automatic page creation from database; enhanced scalability and amplified product creation efficiency by 400% over manual page inputting.
- Implemented payment pipeline with Stripe API and email order confirmation, reducing checkout response time by 30%.
- Coordinated with a colleague to enhance online consultation reservation system, integrating voice and video chat functionalities applying **WebSocket** and **WebRTC** technologies, resulting in 60% decrease in confirmation time.

RESEARCH

UCLA-NLP | Mentor: Kai-Wei Chang, Nanyun Peng

Undergrad Researcher

Los Angeles, CA
June 2024 - June 2025

- Developed a comprehensive benchmark for evaluating visual language models (VLMs) on perception and reasoning quality in QA tasks; tested on 26 VLMs to assess self-critique and correction abilities (**CVPR 2025, Accepted [Link](#)**).
- Managed 3 annotators and built a **Python**-based annotation tool; curated over 1,600 fine-grained annotated samples.
- Introduced a novel evaluation metric, VISCore, combining answer accuracy, stepwise critique accuracy, and critique quality to yield more accurate model assessments.
- Implemented an agent-based critique generation method at each reasoning step, improving critique accuracy by 13.5%.

PROJECTS

Ego2Allo - Visual Perspective-Taking - Team Lead

October 2025 - December 2025

- Fine-tuned VLMs for spatial perspective-taking tasks with SFT and RL-GRPO, achieving 10% higher accuracy.
- Incorporated ReAct function calling to agentic pipeline emphasizing structured reasoning, improving accuracy by 3%.

Pittsburgh & CMU QA System - Team Lead

October 2025 - November 2025

- Collaborate in a team of 3 in building an end-to-end RAG QA system with multiple chunking granularities.
- Created a Small-to-Big hybrid retriever that compute weighted fusion of BM25 score on sentence-level chunks and MiniLM-L6-v2 dense cosine similarity on paragraph-level chunks, achieving 75% accuracy under LLM evaluation.

Job-Search - Independent Developer

October 2023 - October 2025

- Initiated and constructed a job search platform, personalizing job recommendations merely based on user views and likes, incorporated Search, Favorite, and Recommendation functionalities in **Java Servlets** with **REST APIs**.
- Designed a recommendation engine with dense-embeddings and **AI**-based keyword extraction to model user interests, integrating Redis as a high-speed vector database decreasing embedding API calls by 80%.
- Deployed on **AWS EC2** virtual machine, combining **Redis** and **Amazon RDS** to cache and persist search and favorite related data, cutting down latency by 90%.

High Performance URL Shortening Web Server - Tech Lead & Contributor

April 2025 - June 2025

- Led a team of 4 in building an async TCP server in **C++** with **Boost.Asio** with test-driven development; deployed on **GCP** using **Docker** and **Compute Engine**; automated CI/CD with Cloud Build for zero-downtime deployment.
- Built application architecture for URL shortening service with **Redis** caching and **PostgreSQL** persistence, leveraging connection pooling to support 1,000+ concurrent users.

SKILLS

Programming Languages: Java, Python, C++, C, SQL, JavaScript, Bash

Frameworks & Libraries: Spark, React, HTML, CSS, Django, Spring Boot, Tomcat, PyTorch, NumPy

Databases & Data Stores: PostgreSQL, MySQL, SQLite, Redis, MongoDB

Cloud Platforms: AWS, Google Cloud Platform (GCP), Microsoft Azure

DevOps & Tools: Docker, Kubernetes, Git, Linux